

Loan Default Risk Analysis

Task 7 – Critical Evaluation of Principal Component Analysis (PCA)
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1. PCA Methodology

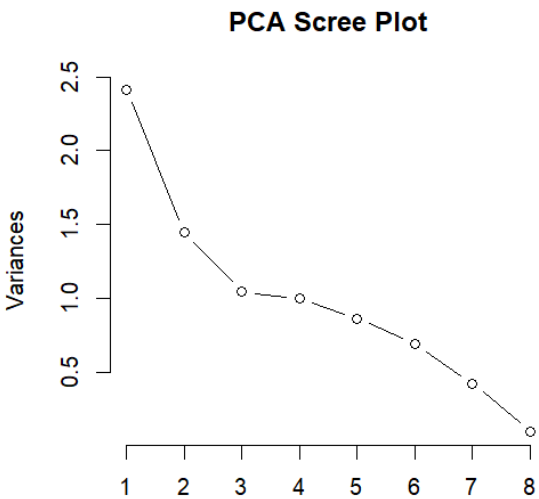
PCA was applied to eight numerical variables: **loan_amount**, **Interest_rate_spread**, **term**, **property_value**, **income**, **Credit_Score**, **LTV** and **dtir1**. The variables **rate_of_interest** and **Upfront_charges** were excluded because Task 5 identified their missingness as a potential data-leakage issue. Rows with missing values were removed, and the variables were standardized using **prcomp(scale. = TRUE)**.

2. PCA Results

Component	Variance Explained	Cumulative Variance
PC1	30.16%	30.16%
PC2	18.09%	48.25%
PC3	13.11%	61.36%
PC4	12.50%	73.86%
PC5	10.84%	84.70%
PC6	8.68%	93.38%
PC7	5.36%	98.73%
PC8	1.27%	100.00%

Key result: The first four principal components explain 73.86% of the total variance, while the first six explain 93.38%.

3. Scree Plot



The scree plot shows a clear reduction in explained variance after the first few components, with PC1 contributing the largest share.

4. Principal Component Loadings

Component	Main variables with larger loadings
PC1	property_value (-0.573), loan_amount (-0.529), income (-0.424), Interest_rate_spread (0.377)
PC2	term (0.589), LTV (0.552), dtir1 (0.456)

5. Is PCA Necessary for This Dataset?

PCA is useful for examining relationships among the numerical variables and for reducing correlated information into a smaller set of components. For this dataset, however, the original feature space contains only eight variables, so dimensionality reduction is not essential for computational reasons. For the complete-case data used in the PCA, **loan_amount** and **property_value** have a correlation of approximately **0.723**, showing meaningful redundancy that PCA can summarize.

6. Advantages and Limitations

Advantages: reduces correlated information, creates uncorrelated components, summarizes variation, and can simplify visualization or later modelling.

Limitations: principal components are combinations of original variables and can be harder to interpret; PCA is unsupervised and does not directly use the default outcome; important predictive information can be lost if too few components are retained.

7. Expected Impact on Predictive Modelling

Using a reduced set of principal components may reduce multicollinearity and produce a more compact feature set. However, because PCA components are less interpretable than the original loan variables, a predictive model using PCA should be compared with a model using the original features. Component selection should also retain enough variance for the intended modelling task.

8. Final Critical Evaluation

For this loan-default dataset, PCA provides a useful exploratory view of the relationships among the eight selected numerical variables. PC1 explains **30.16%** of the variance and is mainly associated with **property_value**, **loan_amount**, **income** and **Interest_rate_spread**. PC2 is mainly associated with **term**, **LTV** and **dtir1**. The first six components retain **93.38%** of the total variance. Therefore, PCA can substantially summarize the numerical information, although its practical benefit is limited by the relatively small number of original variables and the reduced interpretability of transformed components. The final PCA is also aligned with the Task 5 decision to exclude variables affected by the identified data-leakage concern.